

CII National Kaizen Circle Compétition 2019

A Kaizen Presentation By

EVEREST INDUSTRIES LIMITED

(Podanur Works, Coimbatore)

On: 11th & 12th June 2019

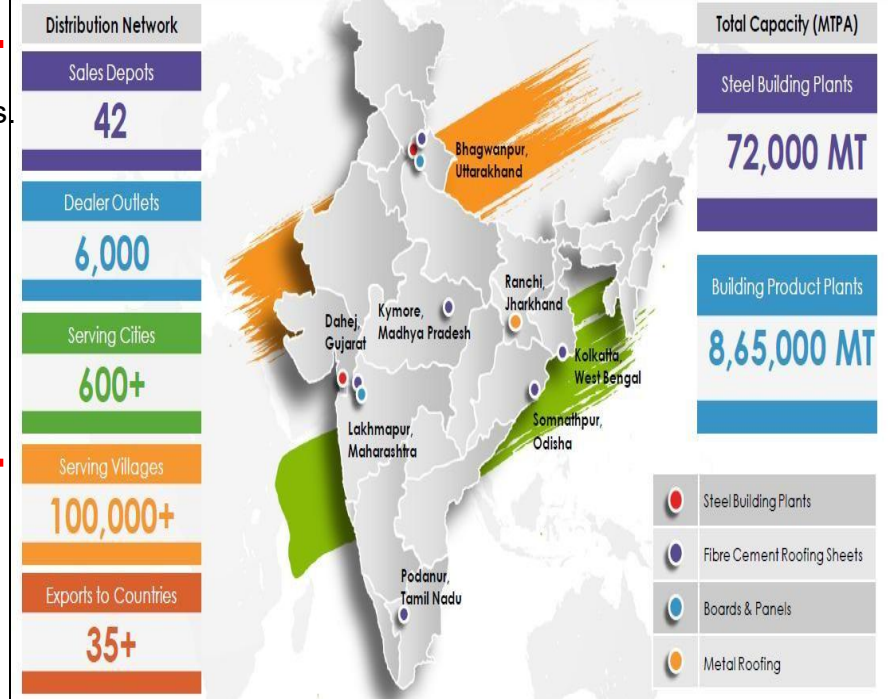
Venue : India Habitat Centre , New Delhi

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

- **Building Products (63%)** – include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Kaizen Theme: To reduce Maintenance spare parts cost of Vacuum Trays.

Implemented Area: Sheeting Machine-1 Shop Floor, Podanur Works, Coimbatore

Implementation Date: 2nd September 2018

Cross Functional Team:



Mr. V. Venkat Rao
(Asst. Manager Maintenance)



Team Leader
Mr. C. Murugesan
(Sr. Manager, Maintenance.)



Mr. K. Karthikeyan
(Sr. Engineer Maintenance)



Mr. M. Vinoth Kumar
(Engineer - Production)



Mr. Y. Kesavaraj
(SM1 Line Incharge)

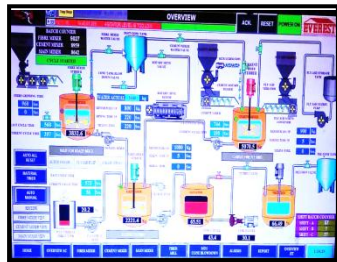


Mr. N. Anil Kumar
Engineer - QS & BE



Mr. Pillai
Fabricator

Process Flow:



**RAW MATERIAL
PREPARATION**

**CONSISTENCY
MIXER**

VATS

**SIEVE
FILTRATION**

**VACUUM
SYSTEM**

DE-PILING

PRE CURING

**SHEET
HANDLING**

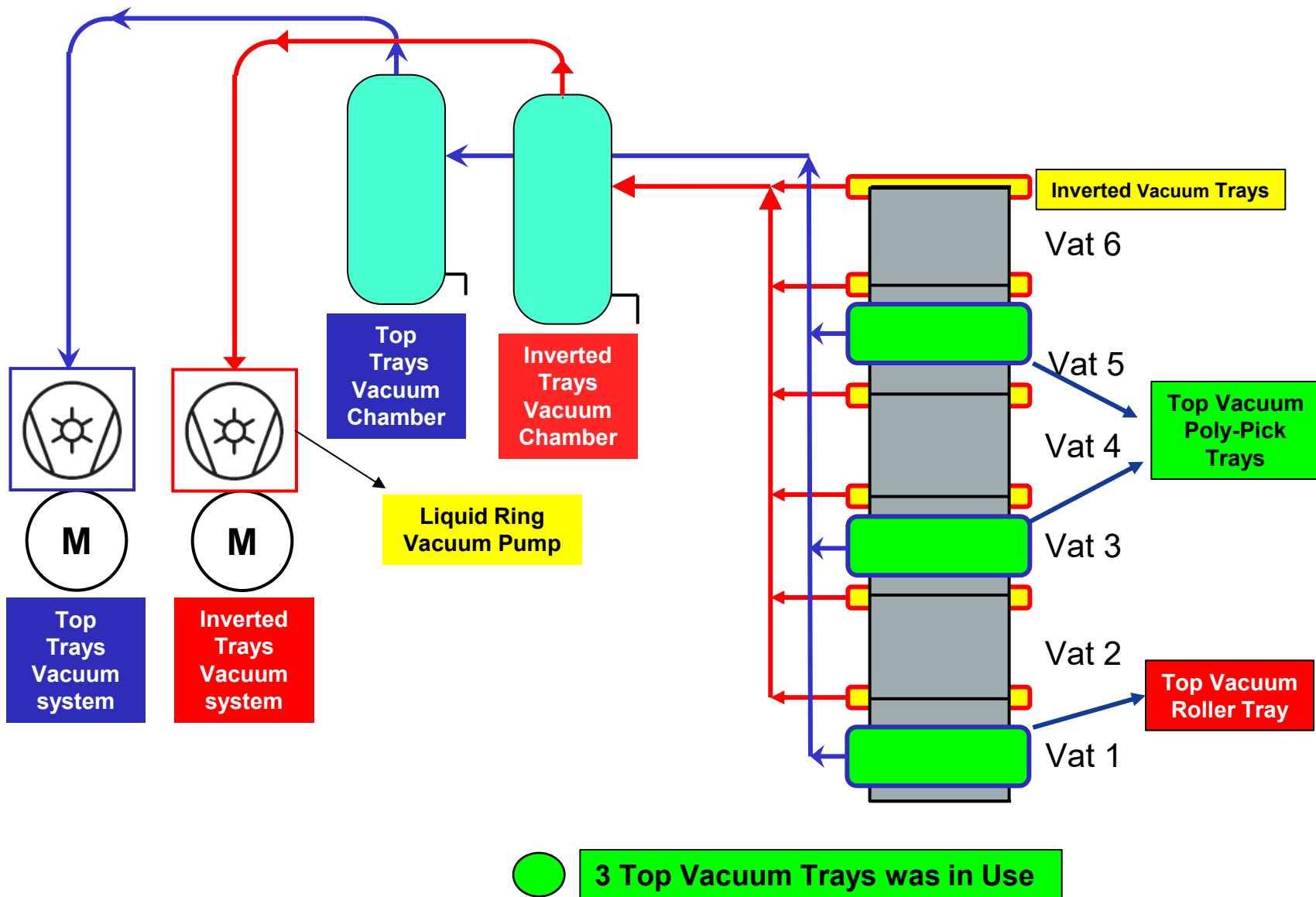
**SHEET
FORMING**

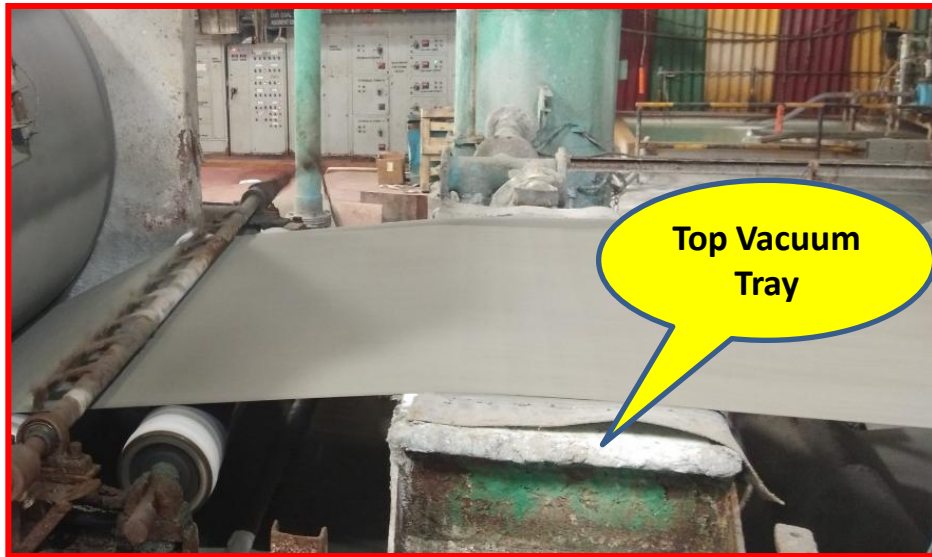


Problem :

More Maintenance cost due to use of Top Vacuum Trays.

Top Vacuum trays consists of Rollers, Bearings, Poly-Picks etc.





- 1 Top vacuum tray consists of 5 roller, with 20 Bearings, Poly-Picks etc which reduces the Load on the Main Drive.
 - Life of Each set Bearings is **15 Days**.
 - Life of Small Rollers is **1 Year**.
 - Life of Poly-Picks is **30 Days**.
 - And regular application of Lubrication to the bearings on **Weekly** basis.

EVEREST Root Cause Analysis & Idea Generation

Problem: More Maintenance Cost in Vacuum System.

Why 1 : Due to Top Vacuum Tray.

Why 2 : Vacuum Tray consists of Maintenance spares like Bearings, Rollers, Poly-Picks & Lubrication which increases the Maintenance cost.

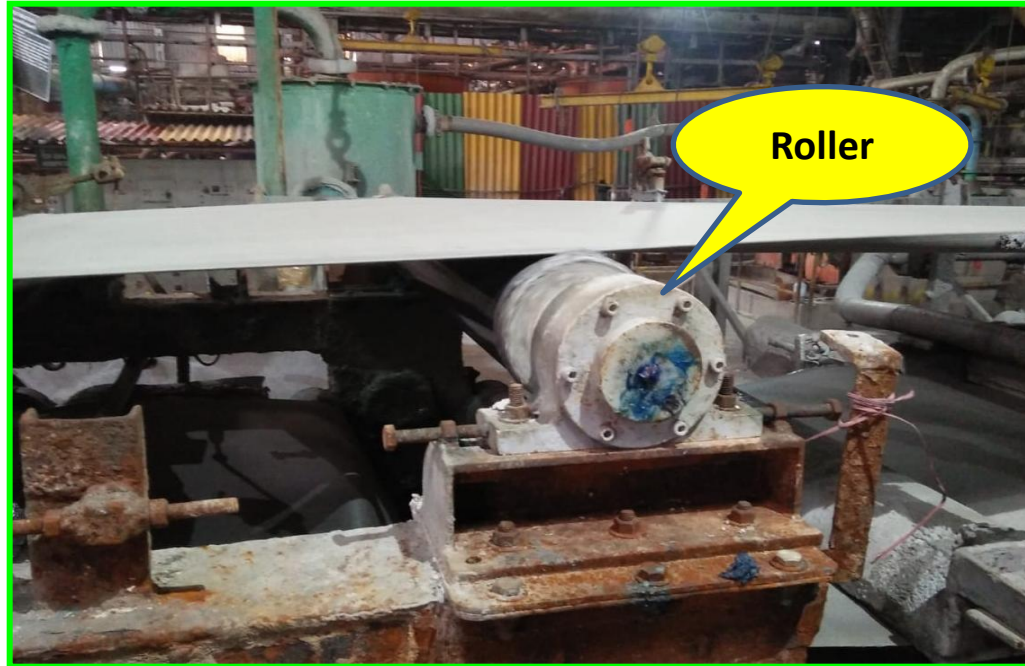
Why 3 : Machine / Design Constraint

Root Cause: Design constraint of the Vacuum Trays system.

Idea Generation: To replace 1 Top Vacuum Trays with a Roller & increasing the vacuum intensity in the other 2 Vacuum Trays.

Action: Replaced 1 Top Vacuum Tray with a Roller.

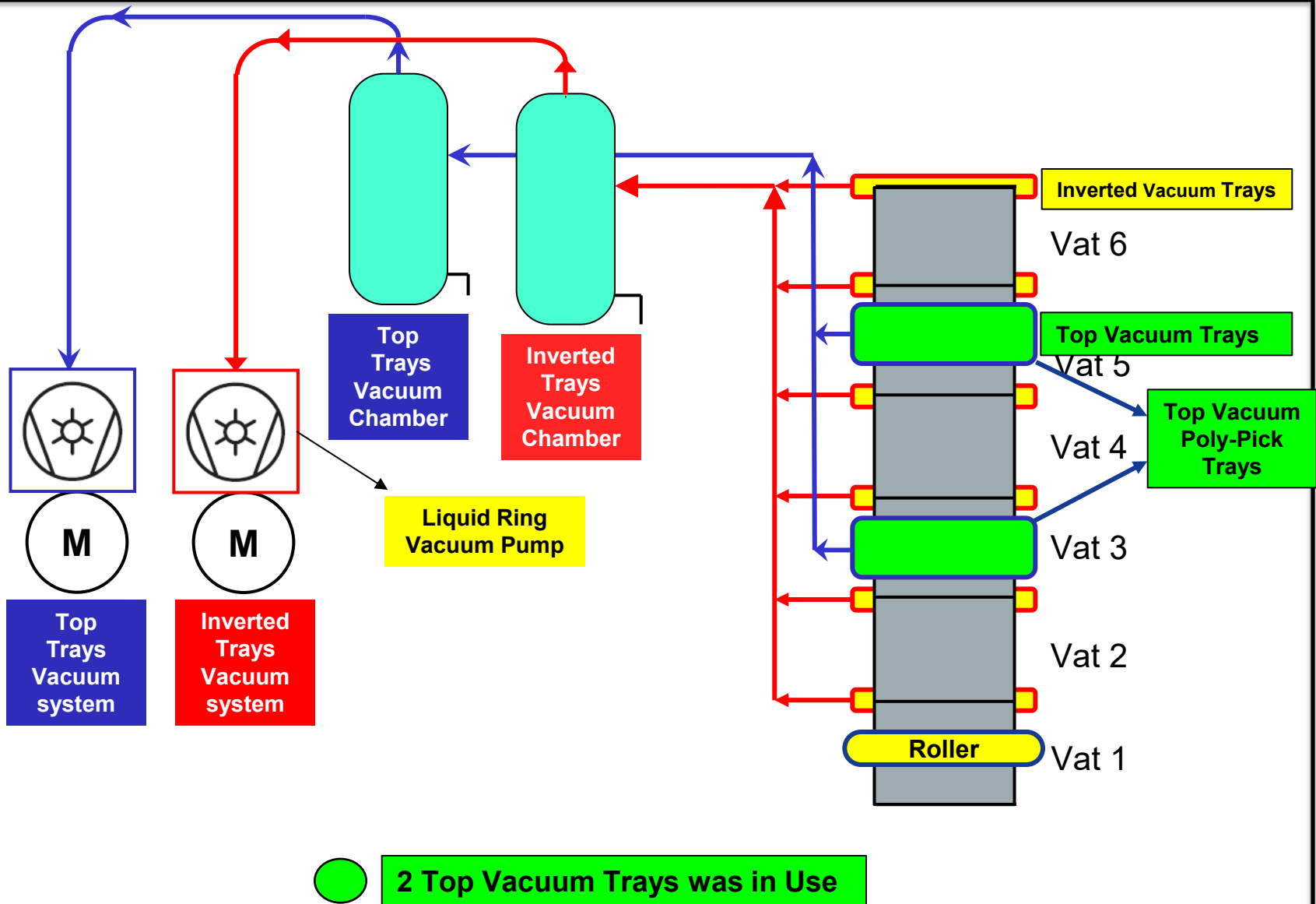
SN	Counter Measure Activities	Target Date	Responsibility	Status
01	Suitable Roller has been arranged from the spares in the factory.	01/09/2018	Maintenance Department	Completed
02	Mounting arrangement has been modified for mounting the roller.	02/09/2018	Maintenance Department	Completed
03	Vacuum intensity has increased for the other 2 Vacuum trays.	02/09/2018	Maintenance Department	Completed



- 1 Top Vacuum Tray is replaced with Roller and increased vacuum intensity of the other 2 Top vacuum trays for vacuum compensation for drying slurry film. By this we eliminated the use of 1 Top vacuum tray which consists of Rollers, Bearings, Poly-Picks etc and reduced maintenance cost.

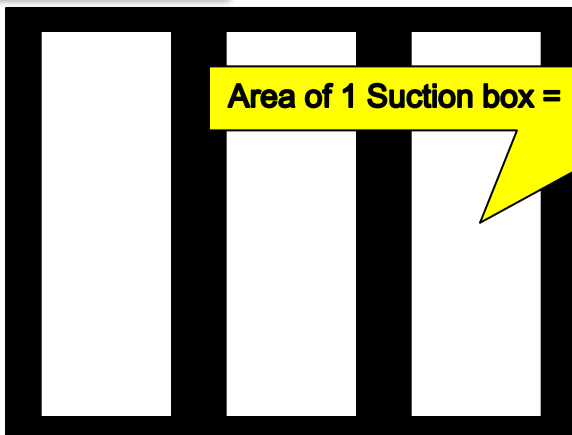


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Before Situation:

Vacuum suction boxes opening Area

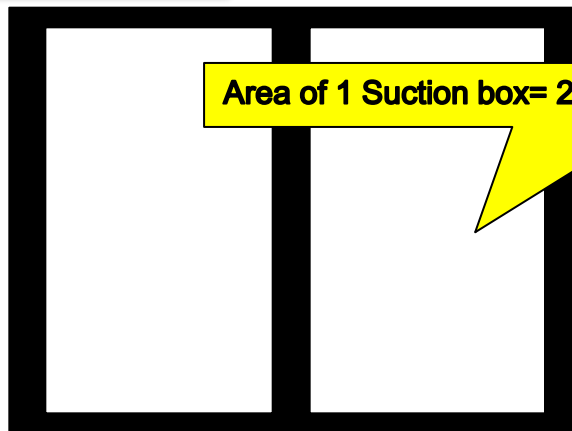


Area of 1 Suction box = 1898 CM²

Top Vacuum Trays



After Situation:



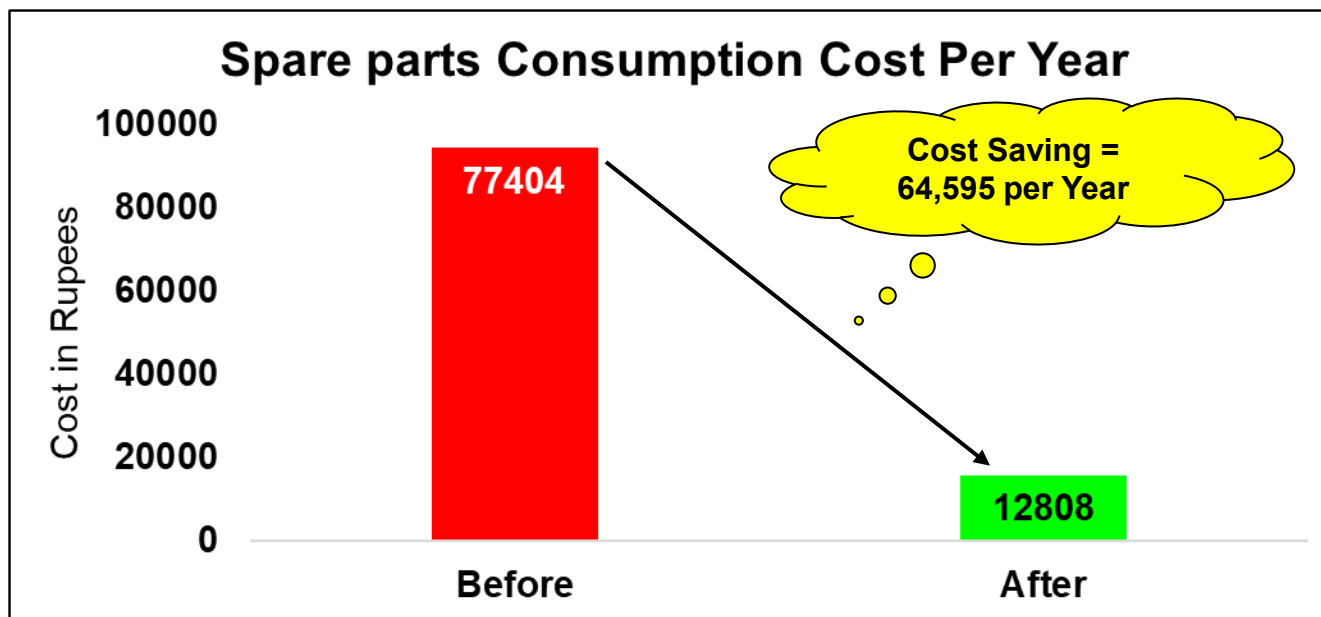
Area of 1 Suction box = 2920 CM²

Top Vacuum Trays



Cost Calculation Per Year

Before			After		
Spares	Quantity	Rs.	Spares	Quantity	Rs.
Small Rollers	5 No's	25641	Bearings	6 No's	5215
Bearings	480 No's	31401	Grease	24 Kgs	7593
Poly-Picks	10 Strips	7315			
Grease	48 Kg	15187			
Total Cost Per Year		77404	Total Cost Per Year		12808



Tangible Benefits

- Use of Rollers, Bearings & Poly-picks in Top vacuum tray is eliminated.
- Potential cost saving of 64,595 Rs./Annum.
- Maintenance Effort Reduced.

In-tangible Benefits

- Morale increased due to reduction in effort for maintenance of Top vacuum tray.

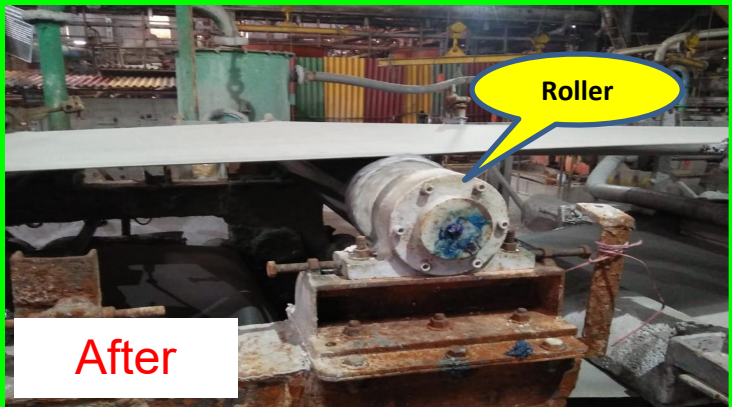
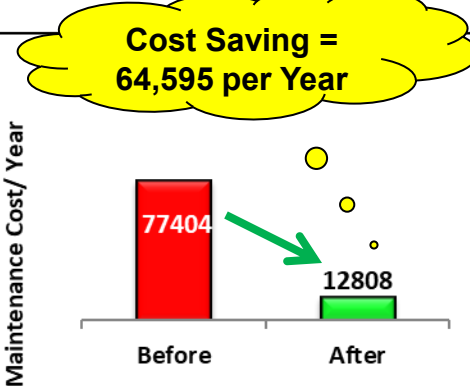
- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Horizontal Deployment:

BW	CW	KW	LW	PW	SW

Kaizen Sheet			Category: 2	Company	MM/YY	S.No												
Reactive	Proactive	Innovative ✓		EVEREST	09/18	3												
Kaizen Title: To reduce Maintenance spare parts cost by replacing one vacuum tray with Roller.					Implemented Area: PW1 Shop floor													
Problem/Present Status: More Maintenance cost due to use of Top vacuum Tray.			 Top Vacuum Tray Before  Roller After		Implemented by: C. Murugesan, V. Venkat Rao, K. Karthikeyan, Kesavaraj, Pillai,													
Root Cause Identification:					Result/Benefit:													
Problem Why 1 Why 2 Why 3 • More maintenance cost. • Due to top Vacuum Tray. • Vacuum trays consists of Maintenance spares like Bearings, rollers. With these maintenance cost is increased. • Design Constraint of system.					(a) Qualitative Reduction of Load on the main drive & Felt adjustment is done easily. (b) Quantitative Cost Saving = 64,595 per Year  Maintenance Cost/ Year Before: 77404, After: 12808													
Root cause:			Design constraint of the system.		Standardization:													
Idea to eliminate root cause:			To replace one Top vacuum tray with roller.		Preventive maintenance as per schedule.													
Counter-measure :			Replaced One Top Vacuum Tray with a Roller. So that, Maintenance cost of the Roller is very less & maintenance effort gets reduced.		<table><tr><td>BW</td><td>CW</td><td>KW</td><td>LW</td><td>PW</td><td>SW</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>		BW	CW	KW	LW	PW	SW						
BW	CW	KW	LW	PW	SW													

Thank You